The EC06 report shows improvement over EC05, particularly through the inclusion of experimental results and integration testing. However, several implementation claims, inconsistencies, and evaluation results require clarification and supporting evidence.

Comments and Suggestions on EC06 Application

1. Functional Requirement Completion (Section 02)

- All 60 functional requirements are marked as Completed, but adequate supporting evidence has not been provided for each functionality.
- Verify the actual implementation status of every requirement and clearly identify any partially implemented or untested functionalities.
- Provide test case references, screenshots, API responses, or other evidence supporting the completion claims.
- Pay particular attention to advanced functionalities such as multiplayer battles, adaptive learning, achievement badges, and administrative features.

2. Changes to the System Design

- The earlier design specified MongoDB and FAISS/ChromaDB, whereas EC06 reports SQLite/PostgreSQL and in-memory vector retrieval.
- Similarly, the previously specified chunk size of 512 tokens has changed to 500 tokens.
- Clearly explain these architectural changes, the reasons for selecting the new technologies, and their implications.
- Ensure the final documentation accurately reflects the implemented system rather than the initially proposed architecture.

3. RAG Implementation and Validation

- Provide technical evidence demonstrating the actual retrieval and question-generation process.
- Include examples of extracted PDF content, generated chunks, retrieved passages, and resulting questions.
- Clearly identify the embedding model, vector representation, similarity measure, and retrieval strategy actually used.
- Demonstrate how generated answers are verified against the uploaded source material.
- Explain how unsupported or incorrect questions are identified and handled.

4. Quiz Generation Performance (Experiment 01)

- The reported average response time of 3.15 seconds is encouraging, but the experiment uses only three runs with a single PDF.
- Extend testing to documents with different sizes, lengths, and content complexity.
- Repeat the experiments under different question counts and system conditions.
- Present individual measurements, processing-stage timings, and supporting logs.
- Clearly identify the hardware, software, and network environment used for testing.

5. Retrieval Relevance Evaluation (Experiment 02)

- The reported retrieval relevance of 90% should be supported by a clearly defined experimental dataset.
- Explain how the 20 retrieved chunks were selected and evaluated.
- Identify the criteria used by the subject reviewer to determine relevance.
- Clarify how the experiment relates to the configured top-K retrieval strategy.
- Include sample retrieval results and consider evaluating multiple PDFs and topics.

6. Generated Question Quality (Experiment 03)
- The reported 92% question acceptance rate is promising, but the evaluation methodology needs further explanation.
- Clearly define the criteria used to assess question correctness, clarity, relevance, and difficulty.
- Explain who evaluated the 50 generated questions and whether the assessment was independently conducted.
- Provide representative examples of accepted and rejected questions.
- Explain how the four questions with ambiguous distractors were corrected or handled.

7. Duplicate Question Detection (Experiment 04)
- The reported detection accuracy of 93.3% is based on only 30 question pairs.
- Expand the test dataset to include paraphrased questions, semantically similar questions, and questions with overlapping terminology but different meanings.
- Report precision, recall, and false-positive/false-negative cases in addition to accuracy.
- Justify the selected 80% similarity threshold.
- Note that the reported 93.3% accuracy is below the earlier proposed target of above 95%; this should be acknowledged and discussed.

8. Adaptive Difficulty Mechanism (Experiment 05)
- The reported 90% classification accuracy requires further explanation, particularly regarding the three unsuccessful scenarios.
- There appears to be an inconsistency between the earlier adaptation thresholds (above 80% and below 60%) and the EC06 thresholds (at least 80% and at most 50%).
- Clarify the exact implemented rules and ensure consistency with the SRS.
- Distinguish between question difficulty classification and learner-performance-based difficulty adjustment, as these are different tasks.
- Demonstrate actual difficulty changes using controlled quiz-attempt histories.
- Explain how the initial difficulty, consecutive attempts, and minimum/maximum difficulty levels are handled.

9. Integration Testing (Section 04)
- All 13 integration tests are marked as accepted, but the document mainly provides brief statements rather than detailed test evidence.
- Include actual test inputs, observed outputs, execution dates, and Pass/Fail results.
- Identify who performed the external-user acceptance tests.
- Provide relevant API logs, database records, or screenshots supporting integration testing.

- Record any failures encountered during testing and explain the corrective actions and retesting performed.

10. Database and Data Management (Section 05)
- Provide the actual database schema or ER diagram, including primary keys, foreign keys, and relationships.
- Explain how document chunks and embeddings are associated with source PDFs and persisted across application restarts.
- Clarify the difference between the development database and the intended production database.
- Verify the claimed compatibility with PostgreSQL through actual testing rather than relying only on ORM support.
- Demonstrate data integrity, user-specific access restrictions, and appropriate deletion of associated records.

11. User Interface and Usability (Section 06)
- The interface design appears reasonably consistent and professional.
- However, most screenshots demonstrate initial interfaces, configuration screens, or empty states rather than completed operations.
- Include screenshots showing successfully uploaded PDFs, generated questions, quiz attempts, results, performance analytics, and adaptive feedback.
- Provide evidence supporting the stated accessibility and responsive-design compliance.
- The usability evaluation involving five students is useful, but individual task completion times, feedback, and identified difficulties should be reported.

12. Multiplayer and Gamification Features
- The report claims completion of multiplayer battles, player matching, leaderboards, badges, and notifications.
- These advanced features require stronger evidence, particularly regarding real-time synchronization and matchmaking.
- Demonstrate a complete multiplayer session using separate user accounts.
- Explain how simultaneous responses, disconnections, timeouts, and score synchronization are handled.
- Support the reported synchronization latency of less than 150 ms with actual measurements.

13. Research Contribution and Comparative Evaluation
- The report mainly evaluates individual system components but provides limited evidence of the effectiveness of the proposed approach compared with existing alternatives.
- Clearly identify the research contribution beyond integrating existing RAG and LLM technologies.
- Consider comparing question quality with and without RAG.
- Evaluate whether adaptive difficulty provides measurable advantages over fixed-difficulty quizzes.

- Discuss the limitations of the experiments and avoid generalizing findings from small test datasets.

14. Consistency and Reliability of Reported Results
- Ensure all reported numerical results are derived from actual experiments and can be reproduced.
- Provide the datasets, evaluation procedures, calculation details, and evidence supporting the reported percentages.
- Explain unsuccessful test cases instead of discussing only successful outcomes.
- Avoid claims such as "fully verified", "reliably adapted", or "meets standards" unless supported by appropriate evidence.
- Resolve inconsistencies between the functional completion claims, experimental results, limitations, and appendix checklists.

15. Documentation and Final Presentation
- Include an updated system architecture diagram and the actual technology stack.
- Improve traceability between SRS requirements, implementation modules, test cases, and results.
- Add properly labelled screenshots and relevant technical diagrams.
- Ensure that the conclusion reflects demonstrated achievements rather than proposed or expected outcomes.
- Remove incomplete, conditional, or template-style statements where actual implementation information should be provided.
- Ensure that all team members can explain their individual contributions, technical decisions, implementation details, and experimental findings.

Overall Recommendation

The EC06 submission is a considerable improvement over EC05 and presents a more complete description of the system and its evaluation. However, the main concern remains the need to substantiate the reported implementation achievements and experimental results with verifiable evidence.

Before final submission, priority should be given to:

1. Verifying all functional completion claims.
2. Resolving inconsistencies in the system architecture and adaptive difficulty rules.
3. Providing reproducible experimental results and supporting evidence.
4. Demonstrating the complete RAG-based adaptive quiz workflow.
5. Strengthening integration testing and documenting unsuccessful test cases.
6. Clarifying and validating the research contribution.